

Executive Summary: Bank Customer Churn Risk Intelligence

For government and public-sector stakeholders

Purpose

This project demonstrates a predictive decision-support prototype for early identification of bank customers who may churn. It is designed to support targeted retention while emphasizing governance, fairness, transparency and human oversight.

Key results

- 10,000 customer records; observed churn rate 20.37%.
- Five supervised models evaluated using a stratified 80/20 holdout set.
- Champion: Gradient Boosting; ROC-AUC 0.869.
- XGBoost accuracy: 87.1%.
- Top predictive signals: age, product count, activity status, balance and geography.
- Dashboard: risk calculator, portfolio analytics, explainability and what-if simulation.

Public-policy relevance

Predictive churn systems can illustrate a transition from broad reactive retention to targeted engagement. For public-sector and supervisory stakeholders, the important lesson is that predictive performance must be accompanied by documented data provenance, validation, threshold governance, subgroup testing, monitoring and human oversight.

Recommended governance principles

- Human oversight for consequential customer actions.
- Transparent model documentation.
- Fairness and subgroup performance testing.
- Periodic validation, drift monitoring and calibration.
- Do not treat predictive association as causal evidence.
- Combine risk with customer value and intervention cost.
- Maintain audit trails for scoring and interventions.

Next steps

- Validate using time-separated bank data.
- Add richer behavioral and service-history variables where appropriate.
- Calibrate probabilities and optimize intervention thresholds.
- Pilot with relationship managers before automation.
- Measure intervention uplift using a suitable control group.

Caveat: This is an educational/project prototype. It should not be used directly for regulatory, lending, pricing or adverse-action decisions without institution-specific validation, compliance review, privacy/security controls and model-risk governance.